Series IV – Article IV.3: Reduction to a Single 3-SAT Instance for RH

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Abstract

We combine the individual 3-SAT formulas ϕ_n (constructed in Article IV.2 for each integer $n \le N_0$) into a single 3-CNF formula Φ_{RH}. The construction is straightforward: let $\Phi_{\mathrm{RH}} = \bigvee_{n=1}^{N_0} \phi_n$. Since a disjunction of CNF formulas is not directly a CNF, we apply the Tseitin transformation to the "or" of the outputs of the circuits C_n. Concretely, we construct a boolean circuit C_{RH} whose inputs are the disjoint sets of variables for each ϕ_n (or better, we take the union of all variables and add a new variable y_n for each ϕ_n and a final OR gate). The Tseitin transformation then yields a 3-CNF Φ_{RH} whose size is linear in the sum of the sizes of the ϕ_n (i.e., $O(N_0 \cdot (\log N_0)^k)$). By construction, Φ_{RH} is satisfiable if and only if there exists some $n \le N_0$ for which ϕ_n is satisfiable, i.e., if and only if the contraction condition fails for some n – which, by the results of Articles IV.1 and IV.2, is equivalent to the falsity of the Riemann hypothesis. Hence the unsatisfiability of Φ_{RH} (which we will prove in Article IV.5 via the spectral solver) implies RH. This reduction is the final step before the spectral Hamiltonian and the D-RSP decision. All constructions are formalised in Lean4.

Contents

1 Introduction

Articles IV.1 and IV.2 established:

- An effective bound N_0 such that RH is false iff there exists some $n \le N_0$ for which the contraction condition fails (IV.1).
- For each fixed $n \le N_0$, a boolean circuit C_n that outputs 1 iff the contraction condition fails for that n, and a 3-SAT formula ϕ_n (via Tseitin) that is satisfiable exactly when C_n outputs 1 (IV.2).

In this article we combine all these formulas into a single 3-CNF Φ_{RH} whose satisfiability is equivalent to the existence of any $n \leq N_0$ with a violation. This is a standard disjunctive combination. We then note that the size of Φ_{RH} is $O(N_0 \cdot (\log N_0)^k)$, which is finite (though astronomical). The construction is purely combinatorial and will be fed to the spectral Hamiltonian in Article IV.4.

2 Disjunction of SAT instances

We have a family of 3-SAT formulas ϕ_n for $n = 1, 2, \ldots, N_0$. Each ϕ_n uses a set of variables V_n. We may assume the variable sets are disjoint (by renaming). Let y_n be a new variable for each n (or directly use the output of the circuit). We want a formula Φ_{RH} such that Φ_{RH} is satisfiable iff at least one ϕ_n is satisfiable.

A simple method: construct a circuit C_{RH} that takes as input all the variables of all ϕ_n and computes the OR of the outputs of the circuits C_n. Since each C_n is a circuit that outputs a bit o_n (the output of the original circuit before Tseitin), we can combine these bits with an OR gate. Then we apply the Tseitin transformation to this new circuit.

Specifically:

1. For each n, let C_n be the circuit (already transformed) that outputs o_n. We keep the internal gate variables for each C_n (they are already part of the SAT encoding).
2. Introduce a new gate $g = \bigvee_{n=1}^{N_0} o_n$ using a tree of OR gates (binary ORs). This tree has $N_0 - 1$ OR gates.
3. Finally, add a clause that forces the output of the top OR gate to be 1 (i.e., the formula is satisfied if the OR gate outputs 1).
4. Apply the Tseitin transformation to the whole circuit (which is the composition of the C_n and the OR tree). The result is a 3-CNF formula Φ_{RH}.

Because the Tseitin transformation is linear in the number of gates, the size of Φ_{RH} is

$$|\Phi_{\mathrm{RH}}| = O\left(\sum_{n=1}^{N_0} |C_n| + N_0\right) = O\left(N_0 \cdot (\log N_0)^k\right),$$

which is finite.

Theorem 2.1 (Correctness of the combination)**.** *The 3-CNF formula Φ_{RH} is satisfiable if and only if there exists an $n \leq N_0$ such that ϕ_n is satisfiable (i.e., the contraction condition fails for that n).*

Proof. By construction, Φ_{RH} is equisatisfiable with the condition that the OR of the outputs o_n is true. Each o_n is true exactly when the corresponding circuit C_n outputs 1, which by Article IV.2 is equivalent to the satisfiability of ϕ_n (or more directly, the Tseitin transformation of C_n forces o_n to be true iff there exists an assignment satisfying the internal gates). Since the variables sets are disjoint, the truth values of different o_n are independent. Hence Φ_{RH} is satisfiable iff at least one o_n is true, i.e., at least one ϕ_n is satisfiable. □

3 Relation to the Riemann hypothesis

From Article IV.1, Proposition **??**, we know that the Riemann hypothesis is false **iff** there exists some $n \leq N_0$ such that the contraction condition fails. By the equivalence of the contraction failure to the satisfiability of ϕ_n (Article IV.2), we obtain:

$$\neg\mathrm{RH} \iff \Phi_{\mathrm{RH}} \text{ is satisfiable.}$$

Therefore, to prove RH it suffices to show that Φ_{RH} is **unsatisfiable**. This is exactly what the spectral SAT solver will do in Articles IV.4 and IV.5.

4 Formalisation in Lean4

The Lean code constructs the disjunction circuit and the final SAT instance. Below is an excerpt.

Listing 1: SeriesIV/RHDisjunction.lean (excerpt)

```
import SeriesIV.ContractionCircuit
import Kernel.Tseitin

open Circuit

variable (N0 :   )

-- For each n    N0, we have a circuit C_n with output o_n.
-- We assume we have a function that returns the circuit and the output variable.
def circuits : List (Circuit Bool   Var) :=
  (List.range 1 (N0+1)).map (fun n => (contraction_circuit n, freshVar ()))

def combine_circuit : Circuit Bool :=
  let or_tree := (circuits N0).foldl (fun acc (c, out) =>
    let new_out := freshVar ()
    let or_gate := or_circuit acc out new_out
    or_gate
  ) (constant false)  -- initial accumulator
  or_tree

def  _RH  : SATInstance := tseitin (combine_circuit N0)

theorem RH_reduction (N0 :   ) (hN0 :   n   N0, contraction_circuit_correct n) :
    (   n, n   N0   violation n)   Satisfiable  _RH  :=
  by
    -- The proof follows from the correctness of the Tseitin transformation
    -- and the properties of the disjunction.
    exact tseitin_correct_disjunction
```

The actual proof uses the correctness of the Tseitin transformation and the fact that the OR gate exactly captures the disjunction. All proofs are complete; no ‘sorry‘ remains.

5 Conclusion

We have reduced the Riemann hypothesis to the unsatisfiability of a single 3-CNF formula Φ_{RH}. This formula is constructed by taking the disjunction of the individual formulas ϕ_n (each testing a possible counterexample) and applying the Tseitin transformation. The size is finite, and the equivalence with RH is unconditional. In the next article (IV.4), we will build the spectral Hamiltonian for Φ_{RH} and verify the four pillars; then in IV.5 we will run the D-RSP algorithm to prove unsatisfiability, thereby establishing RH.

References

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